Experiment 3 – Measuring Central Tendency and Variability

Aim: To calculate mean, median, mode, range, variance, standard deviation and IQR for single and grouped data using Python.

Dataset: `urban_climate.csv`

Keep the CSV file and this notebook in the same folder. Run the cells from top to bottom.

1. Read the dataset

In [1]:
```
import pandas as pd
import numpy as np

df = pd.read_csv("urban_climate.csv")
print("Number of records:", len(df))
df.head()
```

```
Number of records: 11040
```

Out[1]:

	city	country	latitude	longitude	year	month	temperature_celsius	humidity_percent
0	New York	USA	40.7128	-74.006	1980	1	6.21	48.8
1	New York	USA	40.7128	-74.006	1980	2	17.46	38.7
2	New York	USA	40.7128	-74.006	1980	3	20.28	32.3
3	New York	USA	40.7128	-74.006	1980	4	22.31	30.6
4	New York	USA	40.7128	-74.006	1980	5	20.01	22.6

Inference: The dataset is loaded successfully. It contains climate records for different cities.

2. Frequency, mean, median and mode

In [2]:
```
# Frequency of city records
print(df["city"].value_counts())

# Mean, median and mode for three numerical columns
columns = ["temperature_celsius", "humidity_percent", "precipitation_mm"]
```

```
for col in columns:
    print("\n", col)
    print("Frequency:", df[col].count())
    print("Mean:", df[col].mean())
    print("Median:", df[col].median())
    print("Mode:", df[col].mode().tolist())
```

```
city
New York        552
Los Angeles     552
Chicago         552
Houston         552
Phoenix         552
Philadelphia    552
San Antonio     552
San Diego       552
Dallas          552
San Jose        552
London          552
Paris           552
Berlin          552
Tokyo           552
Beijing         552
Mumbai          552
Delhi           552
São Paulo       552
Mexico City     552
Lagos           552
Name: count, dtype: int64

 temperature_celsius
Frequency: 11040
Mean: 13.52115036231884
Median: 13.34
Mode: [6.36, 28.49]

 humidity_percent
Frequency: 11040
Mean: 48.4329438405797
Median: 48.2
Mode: [20.0]

 precipitation_mm
Frequency: 11040
Mean: 66.8118115942029
Median: 60.25
Mode: [44.5]
```

Inference: Mean gives the average, median gives the middle value and mode gives the most repeated value. Frequency shows the number of available records.

3. Manual calculations without statistical inbuilt functions

The following functions use loops and simple calculations. Bubble sort is used to arrange values.

In [3]:
```
# Helper functions for calculating statistics manually.

def manual_sort(values):
    # Merge sort is used to arrange values without using sorted().
    if len(values) <= 1:
        return values[:]

    middle = len(values) // 2
    left = manual_sort(values[:middle])
    right = manual_sort(values[middle:])

    result = []
    i = 0
    j = 0

    while i < len(left) and j < len(right):
        if left[i] <= right[j]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1

    while i < len(left):
        result.append(left[i])
        i += 1

    while j < len(right):
        result.append(right[j])
        j += 1

    return result


def get_mean(values):
    total = 0
    for value in values:
        total += value
    return total / len(values)


def get_median(values):
    data = manual_sort(values)
    n = len(data)
    if n % 2 == 0:
        return (data[n // 2 - 1] + data[n // 2]) / 2
    return data[n // 2]


def get_mode(values):
    frequency = {}
    for value in values:
```

```
        frequency[value] = frequency.get(value, 0) + 1

    highest = max(frequency.values())
    if highest == 1:
        return "No mode"

    modes = []
    for value in frequency:
        if frequency[value] == highest:
            modes.append(value)
    return modes


def get_variance(values):
    mean = get_mean(values)
    squared_total = 0
    for value in values:
        squared_total += (value - mean) ** 2
    if len(values) < 2:
        return 0
    return squared_total / (len(values) - 1)


def get_standard_deviation(values):
    return get_variance(values) ** 0.5


def get_quartiles(values):
    data = manual_sort(values)
    n = len(data)
    q2 = get_median(data)

    if n % 2 == 0:
        lower = data[:n // 2]
        upper = data[n // 2:]
    else:
        lower = data[:n // 2]
        upper = data[n // 2 + 1:]

    q1 = get_median(lower)
    q3 = get_median(upper)
    return q1, q2, q3, q3 - q1


def read_numeric_column(rows, column):
    values = []
    for row in rows:
        try:
            values.append(float(row[column]))
        except (ValueError, TypeError, KeyError):
            pass
    return values
```

In [4]:

```
# Use the manual functions on the temperature column
values = df["temperature_celsius"].dropna().tolist()
```

```
print("Mean:", get_mean(values))
print("Median:", get_median(values))
print("Mode:", get_mode(values))
print("Sample variance:", get_variance(values))
print("Sample standard deviation:", get_standard_deviation(values))
print("Q1, Q2, Q3 and IQR:", get_quartiles(values))
```

```
Mean: 13.521150362318856
Median: 13.34
Mode: [28.49, 6.36]
Sample variance: 193.69927167228963
Sample standard deviation: 13.917588572460735
Q1, Q2, Q3 and IQR: (2.81, 13.34, 23.85, 21.040000000000003)
```

Inference: The statistics are calculated manually using loops and formulas rather than ready-made statistical functions.

4. Variance and standard deviation for multiple attributes

In [5]:
```
for col in ["temperature_celsius", "humidity_percent", "precipitation_mm"]:
    values = df[col].dropna().tolist()
    print("\nAttribute:", col)
    print("Variance:", get_variance(values))
    print("Standard deviation:", get_standard_deviation(values))
```

```
Attribute: temperature_celsius
Variance: 193.69927167228963
Standard deviation: 13.917588572460735

Attribute: humidity_percent
Variance: 163.64589259219
Standard deviation: 12.792415432286038

Attribute: precipitation_mm
Variance: 1061.7614819972923
Standard deviation: 32.58468170777938
```

Inference: Variance and standard deviation describe how far the values are spread around the mean. A larger value indicates greater variation.

5. Quartiles and IQR for multiple attributes

In [6]:
```
for col in ["temperature_celsius", "humidity_percent", "precipitation_mm"]:
    values = df[col].dropna().tolist()
    q1, q2, q3, iqr = get_quartiles(values)
    print("\nAttribute:", col)
    print("Q1:", q1)
    print("Median (Q2):", q2)
    print("Q3:", q3)
    print("IQR:", iqr)
```

```
Attribute: temperature_celsius
Q1: 2.81
Median (Q2): 13.34
Q3: 23.85
IQR: 21.040000000000003

Attribute: humidity_percent
Q1: 39.4
Median (Q2): 48.2
Q3: 57.5
IQR: 18.1

Attribute: precipitation_mm
Q1: 41.7
Median (Q2): 60.25
Q3: 86.95
IQR: 45.25
```

Inference: IQR is the difference between Q3 and Q1. It describes the spread of the middle 50% of the data.

6. Grouped data analysis by city

In [7]:
```python
for city, group in df.groupby("city"):
    print("\nCity:", city)
    print("Frequency:", len(group))

    for col in ["temperature_celsius", "humidity_percent", "precipitation_mm"]:
        values = group[col].dropna().tolist()
        q1, q2, q3, iqr = get_quartiles(values)

        print("\n", col)
        print("Mean:", get_mean(values))
        print("Mode:", get_mode(values))
        print("Median:", get_median(values))
        print("Variance:", get_variance(values))
        print("Standard deviation:", get_standard_deviation(values))
        print("IQR:", iqr)
```

```
City: Beijing
Frequency: 552

 temperature_celsius
Mean: 9.565978260869565
Mode: [22.04, -3.34, 3.4]
Median: 9.51
Variance: 118.26472426812909
Standard deviation: 10.874958586961565
IQR: 20.69

 humidity_percent
Mean: 54.96376811594205
Mode: [50.6]
Median: 54.9
Variance: 88.5880859570215
Standard deviation: 9.41212441253416
IQR: 12.450000000000003

 precipitation_mm
Mean: 58.66775362318839
Mode: [23.1, 49.7, 52.5, 43.6, 39.4]
Median: 54.45
Variance: 662.8117350535265
Standard deviation: 25.74513031727605
IQR: 35.300000000000004

City: Berlin
Frequency: 552

 temperature_celsius
Mean: -1.594257246376813
Mode: [-3.15, -16.25]
Median: -2.045
Variance: 116.27584663424857
Standard deviation: 10.783127868770201
IQR: 19.95

 humidity_percent
Mean: 60.351811594202864
Mode: [51.9, 60.2]
Median: 60.6
Variance: 92.47506023304139
Standard deviation: 9.616395386684212
IQR: 13.850000000000009

 precipitation_mm
Mean: 59.40688405797103
Mode: [47.4]
Median: 52.650000000000006
Variance: 759.0351612351712
Standard deviation: 27.550592756511996
IQR: 40.19999999999999

City: Chicago
Frequency: 552
```

```
 temperature_celsius
Mean: 6.774528985507244
Mode: [18.33, 11.52, 0.39, -2.03, 13.62, -4.59, 1.44, 20.2, 20.46, 6.99, -8.45, 16.2
7, -1.98, 5.63, 6.66, 21.24, -7.79, -7.09, 16.71, 21.39, 18.69, 16.2, -6.73, -4.27,
-3.39, -5.14, -5.16, -3.07, -2.01, 13.21, -1.63, -7.99, 12.26, 19.6, 22.36, 19.78, -
0.26, -5.17, -1.54, 16.36]
Median: 6.875
Variance: 118.36361937846857
Standard deviation: 10.87950455574465
IQR: 20.045

 humidity_percent
Mean: 38.92065217391303
Mode: [20.0]
Median: 38.650000000000006
Variance: 85.24643296772668
Standard deviation: 9.232899488661548
IQR: 12.900000000000006

 precipitation_mm
Mean: 58.589492753623176
Mode: [39.3, 49.2]
Median: 51.849999999999994
Variance: 787.006568163287
Standard deviation: 28.053637342834655
IQR: 41.55

City: Dallas
Frequency: 552

 temperature_celsius
Mean: 14.339257246376802
Mode: [27.46, 2.07]
Median: 13.835
Variance: 116.09489327671687
Standard deviation: 10.774734023479043
IQR: 20.295

 humidity_percent
Mean: 59.26268115942031
Mode: [61.4, 62.6]
Median: 59.650000000000006
Variance: 90.30078263499838
Standard deviation: 9.50267239438456
IQR: 13.799999999999997

 precipitation_mm
Mean: 60.56413043478257
Mode: [44.0, 48.3, 38.4, 52.2, 59.3, 27.7]
Median: 52.6
Variance: 858.6701810936642
Standard deviation: 29.30307460137356
IQR: 45.95

City: Delhi
```

```
Frequency: 552

 temperature_celsius
Mean: 18.29916666666668
Mode: [8.26, 3.87, 5.17, 28.72]
Median: 18.725
Variance: 113.82331037507572
Standard deviation: 10.66880079367291
IQR: 20.115000000000002

 humidity_percent
Mean: 38.48985507246379
Mode: [20.0]
Median: 38.6
Variance: 86.90138509692524
Standard deviation: 9.322091240538533
IQR: 13.850000000000005

 precipitation_mm
Mean: 80.06938405797105
Mode: [48.8]
Median: 74.95
Variance: 1219.6671371879847
Standard deviation: 34.92373315079568
IQR: 50.4

City: Houston
Frequency: 552

 temperature_celsius
Mean: 16.58789855072463
Mode: [3.39, 25.42, 16.84]
Median: 16.84
Variance: 116.24984784449875
Standard deviation: 10.781922270379189
IQR: 20.3

 humidity_percent
Mean: 47.35760869565221
Mode: [46.3]
Median: 47.3
Variance: 101.43511441647593
Standard deviation: 10.07150010755478
IQR: 13.950000000000003

 precipitation_mm
Mean: 79.91648550724636
Mode: [48.3, 57.8, 60.3]
Median: 72.75
Variance: 1346.3499092230181
Standard deviation: 36.692641077238065
IQR: 53.05

City: Lagos
Frequency: 552
```

```
 temperature_celsius
Mean: 35.14650362318841
Mode: [48.02, 44.95]
Median: 35.355000000000004
Variance: 114.63171642829906
Standard deviation: 10.706620215002447
IQR: 20.175000000000004

 humidity_percent
Mean: 62.91865942028989
Mode: [61.7]
Median: 62.6
Variance: 94.09542251900379
Standard deviation: 9.700279507261829
IQR: 14.249999999999993

 precipitation_mm
Mean: 81.44130434782612
Mode: [52.8, 76.7, 48.7]
Median: 75.05
Variance: 1426.3824650832491
Standard deviation: 37.76747893470319
IQR: 53.800000000000004

City: London
Frequency: 552

 temperature_celsius
Mean: -0.8286231884057969
Mode: [-1.0, 14.56, -15.55, -15.79]
Median: -0.74
Variance: 116.89140281964286
Standard deviation: 10.811632754567778
IQR: 20.525000000000002

 humidity_percent
Mean: 52.751992753623206
Mode: [48.1, 56.2, 50.8]
Median: 52.45
Variance: 94.80946970593664
Standard deviation: 9.737015441393561
IQR: 12.800000000000004

 precipitation_mm
Mean: 58.969927536231914
Mode: [39.9]
Median: 52.9
Variance: 709.2240304847579
Standard deviation: 26.63126039985261
IQR: 38.60000000000001

City: Los Angeles
Frequency: 552

 temperature_celsius
Mean: 14.142010869565224
```

```
Mode: [14.08, 3.24]
Median: 13.92
Variance: 117.36278596721372
Standard deviation: 10.833410634108436
IQR: 20.255000000000003

 humidity_percent
Mean: 46.11304347826084
Mode: [47.9]
Median: 45.8
Variance: 87.10197111970345
Standard deviation: 9.332843678092088
IQR: 12.600000000000001

 precipitation_mm
Mean: 60.06286231884059
Mode: [48.2, 66.3, 63.4]
Median: 52.95
Variance: 829.1971845327338
Standard deviation: 28.795784145126763
IQR: 42.349999999999994

City: Mexico City
Frequency: 552

 temperature_celsius
Mean: 24.78173913043475
Mode: [25.07]
Median: 25.07
Variance: 115.74164669770367
Standard deviation: 10.758329177790744
IQR: 20.145000000000003

 humidity_percent
Mean: 54.24547101449278
Mode: [58.3, 48.4, 49.3, 58.2, 53.5, 51.4, 49.7, 57.3, 65.2]
Median: 54.05
Variance: 94.95740231857734
Standard deviation: 9.744608884843831
IQR: 13.450000000000003

 precipitation_mm
Mean: 78.6400362318841
Mode: [50.6]
Median: 72.30000000000001
Variance: 1290.4455992727326
Standard deviation: 35.92277271136977
IQR: 52.69999999999999

City: Mumbai
Frequency: 552

 temperature_celsius
Mean: 25.28670289855072
Mode: [11.26, 10.1]
Median: 25.375
```

```
Variance: 115.17574627817676
Standard deviation: 10.731996378967743
IQR: 20.084999999999997

 humidity_percent
Mean: 35.54347826086958
Mode: [20.0]
Median: 34.9
Variance: 87.33545648228512
Standard deviation: 9.345344107216444
IQR: 13.349999999999998

 precipitation_mm
Mean: 80.96141304347833
Mode: [56.1, 94.3, 62.5, 64.2, 61.1, 88.9, 45.0, 68.3, 94.7, 53.7, 80.8]
Median: 74.05
Variance: 1406.8797606131159
Standard deviation: 37.50839586830015
IQR: 55.80000000000001

City: New York
Frequency: 552

 temperature_celsius
Mean: 8.812753623188408
Mode: [-2.66]
Median: 9.15
Variance: 114.9340711696782
Standard deviation: 10.720730906504379
IQR: 20.325

 humidity_percent
Mean: 46.655434782608715
Mode: [41.4]
Median: 46.8
Variance: 85.31118638049396
Standard deviation: 9.236405490259399
IQR: 12.5

 precipitation_mm
Mean: 59.96086956521741
Mode: [37.1, 38.0, 43.8, 33.4, 49.1, 45.6]
Median: 53.849999999999994
Variance: 761.7005712933003
Standard deviation: 27.598923371995877
IQR: 40.75

City: Paris
Frequency: 552

 temperature_celsius
Mean: 1.2631702898550745
Mode: [15.24, 0.33, -12.81]
Median: 0.835
Variance: 117.41714365152951
Standard deviation: 10.835919141980042
```

```
IQR: 20.605

 humidity_percent
Mean: 53.68659420289855
Mode: [54.4]
Median: 53.75
Variance: 84.96493792577385
Standard deviation: 9.21764275320832
IQR: 12.5

 precipitation_mm
Mean: 60.13260869565213
Mode: [26.1, 47.5, 41.1]
Median: 53.1
Variance: 771.0053957231916
Standard deviation: 27.76698391477172
IQR: 40.75

City: Philadelphia
Frequency: 552

 temperature_celsius
Mean: 8.371431159420295
Mode: [17.56]
Median: 8.655000000000001
Variance: 117.00384059779324
Standard deviation: 10.816831356630889
IQR: 21.005

 humidity_percent
Mean: 48.29601449275356
Mode: [51.0]
Median: 48.2
Variance: 88.1399840869038
Standard deviation: 9.38828973172983
IQR: 13.699999999999996

 precipitation_mm
Mean: 59.509420289855065
Mode: [40.4, 46.4]
Median: 51.9
Variance: 767.2237586469919
Standard deviation: 27.69880428189982
IQR: 39.25

City: Phoenix
Frequency: 552

 temperature_celsius
Mean: 13.50211956521739
Mode: [27.46, 12.09, 26.74]
Median: 12.945
Variance: 117.37266755010651
Standard deviation: 10.833866694311247
IQR: 20.365
```

```
 humidity_percent
Mean: 49.64891304347832
Mode: [41.0, 47.8, 47.4]
Median: 49.150000000000006
Variance: 95.21745798153577
Standard deviation: 9.757943327440254
IQR: 13.349999999999994

 precipitation_mm
Mean: 61.00072463768115
Mode: [63.5, 49.1]
Median: 55.849999999999994
Variance: 763.7270049185926
Standard deviation: 27.63561117324154
IQR: 39.05

City: San Antonio
Frequency: 552

 temperature_celsius
Mean: 16.81012681159421
Mode: [2.18]
Median: 16.805
Variance: 115.4190742125647
Standard deviation: 10.743326962005982
IQR: 20.735

 humidity_percent
Mean: 61.81268115942027
Mode: [59.2]
Median: 62.0
Variance: 88.56459388726691
Standard deviation: 9.410876361278312
IQR: 13.5

 precipitation_mm
Mean: 80.39257246376823
Mode: [82.0, 83.9, 65.4, 30.5]
Median: 73.85
Variance: 1307.7043911925603
Standard deviation: 36.16219560801805
IQR: 55.35000000000001

City: San Diego
Frequency: 552

 temperature_celsius
Mean: 14.1866847826087
Mode: [29.52, -0.87, -0.8, 14.7]
Median: 14.219999999999999
Variance: 120.18730314546671
Standard deviation: 10.962996996509062
IQR: 20.82

 humidity_percent
Mean: 40.57228260869566
```

```
Mode: [20.0]
Median: 40.8
Variance: 85.53580022488751
Standard deviation: 9.248556656305215
IQR: 12.700000000000003

 precipitation_mm
Mean: 59.71467391304344
Mode: [44.5]
Median: 54.150000000000006
Variance: 743.883250710171
Standard deviation: 27.274223191690922
IQR: 40.64999999999999

City: San Jose
Frequency: 552

 temperature_celsius
Mean: 10.5841304347826
Mode: [-2.77, -1.59]
Median: 10.895
Variance: 117.7295506036455
Standard deviation: 10.850324907745643
IQR: 20.575

 humidity_percent
Mean: 35.9494565217391
Mode: [20.0]
Median: 35.849999999999994
Variance: 90.20653327941292
Standard deviation: 9.497712002341034
IQR: 13.199999999999996

 precipitation_mm
Mean: 60.277173913043455
Mode: [30.6]
Median: 53.849999999999994
Variance: 776.3841967174301
Standard deviation: 27.86367163023262
IQR: 41.45

City: São Paulo
Frequency: 552

 temperature_celsius
Mean: 21.60260869565218
Mode: [21.17, 33.98, 34.7, 35.62]
Median: 21.79
Variance: 116.34985452536903
Standard deviation: 10.786558975195428
IQR: 20.425

 humidity_percent
Mean: 38.638586956521756
Mode: [20.0]
Median: 38.95
```

```
Variance: 74.95852649333233
Standard deviation: 8.657859232704833
IQR: 12.049999999999997

 precipitation_mm
Mean: 78.86322463768109
Mode: [58.7]
Median: 69.1
Variance: 1415.7654146282118
Standard deviation: 37.62665829738554
IQR: 53.85000000000001

City: Tokyo
Frequency: 552

 temperature_celsius
Mean: 12.789076086956522
Mode: [22.69, 1.19, 6.36, 0.23]
Median: 12.815
Variance: 116.16382001597887
Standard deviation: 10.777932084401852
IQR: 19.955000000000002

 humidity_percent
Mean: 42.47989130434786
Mode: [47.3, 43.7, 47.8, 45.0]
Median: 42.8
Variance: 93.15126459796411
Standard deviation: 9.651490278602788
IQR: 13.299999999999997

 precipitation_mm
Mean: 59.095289855072494
Mode: [43.7, 44.7, 23.1]
Median: 53.55
Variance: 767.9268017307148
Standard deviation: 27.711492232117614
IQR: 40.55

 humidity_percent
Mean: 60.351811594202864
Mode: [51.9, 60.2]
Median: 60.6
Variance: 92.47506023304139
Standard deviation: 9.616395386684212
IQR: 13.850000000000009

 precipitation_mm
Mean: 59.40688405797103
Mode: [47.4]
Median: 52.650000000000006
Variance: 759.0351612351712
Standard deviation: 27.550592756511996
IQR: 40.19999999999999

City: Chicago
```

```
Frequency: 552

 temperature_celsius
Mean: 6.774528985507244
Mode: [18.33, 11.52, 0.39, -2.03, 13.62, -4.59, 1.44, 20.2, 20.46, 6.99, -8.45, 16.2
7, -1.98, 5.63, 6.66, 21.24, -7.79, -7.09, 16.71, 21.39, 18.69, 16.2, -6.73, -4.27,
-3.39, -5.14, -5.16, -3.07, -2.01, 13.21, -1.63, -7.99, 12.26, 19.6, 22.36, 19.78, -
0.26, -5.17, -1.54, 16.36]
Median: 6.875
Variance: 118.36361937846857
Standard deviation: 10.87950455574465
IQR: 20.045

 humidity_percent
Mean: 38.92065217391303
Mode: [20.0]
Median: 38.650000000000006
Variance: 85.24643296772668
Standard deviation: 9.232899488661548
IQR: 12.900000000000006

 precipitation_mm
Mean: 58.589492753623176
Mode: [39.3, 49.2]
Median: 51.849999999999994
Variance: 787.006568163287
Standard deviation: 28.053637342834655
IQR: 41.55

City: Dallas
Frequency: 552

 temperature_celsius
Mean: 14.339257246376802
Mode: [27.46, 2.07]
Median: 13.835
Variance: 116.09489327671687
Standard deviation: 10.774734023479043
IQR: 20.295

 humidity_percent
Mean: 59.26268115942031
Mode: [61.4, 62.6]
Median: 59.650000000000006
Variance: 90.30078263499838
Standard deviation: 9.50267239438456
IQR: 13.799999999999997

 precipitation_mm
Mean: 60.56413043478257
Mode: [44.0, 48.3, 38.4, 52.2, 59.3, 27.7]
Median: 52.6
Variance: 858.6701810936642
Standard deviation: 29.30307460137356
IQR: 45.95
```

```
City: Delhi
Frequency: 552

 temperature_celsius
Mean: 18.29916666666668
Mode: [8.26, 3.87, 5.17, 28.72]
Median: 18.725
Variance: 113.82331037507572
Standard deviation: 10.66880079367291
IQR: 20.115000000000002

 humidity_percent
Mean: 38.48985507246379
Mode: [20.0]
Median: 38.6
Variance: 86.90138509692524
Standard deviation: 9.322091240538533
IQR: 13.850000000000005

 precipitation_mm
Mean: 80.06938405797105
Mode: [48.8]
Median: 74.95
Variance: 1219.6671371879847
Standard deviation: 34.92373315079568
IQR: 50.4

City: Houston
Frequency: 552

 temperature_celsius
Mean: 16.58789855072463
Mode: [3.39, 25.42, 16.84]
Median: 16.84
Variance: 116.24984784449875
Standard deviation: 10.781922270379189
IQR: 20.3

 humidity_percent
Mean: 47.35760869565221
Mode: [46.3]
Median: 47.3
Variance: 101.43511441647593
Standard deviation: 10.07150010755478
IQR: 13.950000000000003

 precipitation_mm
Mean: 79.91648550724636
Mode: [48.3, 57.8, 60.3]
Median: 72.75
Variance: 1346.3499092230181
Standard deviation: 36.692641077238065
IQR: 53.05

City: Lagos
Frequency: 552
```

```
 temperature_celsius
Mean: 35.14650362318841
Mode: [48.02, 44.95]
Median: 35.355000000000004
Variance: 114.63171642829906
Standard deviation: 10.706620215002447
IQR: 20.175000000000004

 humidity_percent
Mean: 62.91865942028989
Mode: [61.7]
Median: 62.6
Variance: 94.09542251900379
Standard deviation: 9.700279507261829
IQR: 14.249999999999993

 precipitation_mm
Mean: 81.44130434782612
Mode: [52.8, 76.7, 48.7]
Median: 75.05
Variance: 1426.3824650832491
Standard deviation: 37.76747893470319
IQR: 53.800000000000004

City: London
Frequency: 552

 temperature_celsius
Mean: -0.8286231884057969
Mode: [-1.0, 14.56, -15.55, -15.79]
Median: -0.74
Variance: 116.89140281964286
Standard deviation: 10.811632754567778
IQR: 20.525000000000002

 humidity_percent
Mean: 52.751992753623206
Mode: [48.1, 56.2, 50.8]
Median: 52.45
Variance: 94.80946970593664
Standard deviation: 9.737015441393561
IQR: 12.800000000000004

 precipitation_mm
Mean: 58.969927536231914
Mode: [39.9]
Median: 52.9
Variance: 709.2240304847579
Standard deviation: 26.63126039985261
IQR: 38.60000000000001

City: Los Angeles
Frequency: 552

 temperature_celsius
```

```
Mean: 14.142010869565224
Mode: [14.08, 3.24]
Median: 13.92
Variance: 117.36278596721372
Standard deviation: 10.833410634108436
IQR: 20.255000000000003

 humidity_percent
Mean: 46.11304347826084
Mode: [47.9]
Median: 45.8
Variance: 87.10197111970345
Standard deviation: 9.332843678092088
IQR: 12.600000000000001

 precipitation_mm
Mean: 60.06286231884059
Mode: [48.2, 66.3, 63.4]
Median: 52.95
Variance: 829.1971845327338
Standard deviation: 28.795784145126763
IQR: 42.349999999999994

City: Mexico City
Frequency: 552

 temperature_celsius
Mean: 24.78173913043475
Mode: [25.07]
Median: 25.07
Variance: 115.74164669770367
Standard deviation: 10.758329177790744
IQR: 20.145000000000003

 humidity_percent
Mean: 54.24547101449278
Mode: [58.3, 48.4, 49.3, 58.2, 53.5, 51.4, 49.7, 57.3, 65.2]
Median: 54.05
Variance: 94.95740231857734
Standard deviation: 9.744608884843831
IQR: 13.450000000000003

 precipitation_mm
Mean: 78.6400362318841
Mode: [50.6]
Median: 72.30000000000001
Variance: 1290.4455992727326
Standard deviation: 35.92277271136977
IQR: 52.69999999999999

City: Mumbai
Frequency: 552

 temperature_celsius
Mean: 25.28670289855072
Mode: [11.26, 10.1]
```

```
Median: 25.375
Variance: 115.17574627817676
Standard deviation: 10.731996378967743
IQR: 20.084999999999997

 humidity_percent
Mean: 35.54347826086958
Mode: [20.0]
Median: 34.9
Variance: 87.33545648228512
Standard deviation: 9.345344107216444
IQR: 13.349999999999998

 precipitation_mm
Mean: 80.96141304347833
Mode: [56.1, 94.3, 62.5, 64.2, 61.1, 88.9, 45.0, 68.3, 94.7, 53.7, 80.8]
Median: 74.05
Variance: 1406.8797606131159
Standard deviation: 37.50839586830015
IQR: 55.80000000000001

City: New York
Frequency: 552

 temperature_celsius
Mean: 8.812753623188408
Mode: [-2.66]
Median: 9.15
Variance: 114.9340711696782
Standard deviation: 10.720730906504379
IQR: 20.325

 humidity_percent
Mean: 46.655434782608715
Mode: [41.4]
Median: 46.8
Variance: 85.31118638049396
Standard deviation: 9.236405490259399
IQR: 12.5

 precipitation_mm
Mean: 59.96086956521741
Mode: [37.1, 38.0, 43.8, 33.4, 49.1, 45.6]
Median: 53.849999999999994
Variance: 761.7005712933003
Standard deviation: 27.598923371995877
IQR: 40.75

City: Paris
Frequency: 552

 temperature_celsius
Mean: 1.2631702898550745
Mode: [15.24, 0.33, -12.81]
Median: 0.835
Variance: 117.41714365152951
```

```
Standard deviation: 10.835919141980042
IQR: 20.605

 humidity_percent
Mean: 53.68659420289855
Mode: [54.4]
Median: 53.75
Variance: 84.96493792577385
Standard deviation: 9.21764275320832
IQR: 12.5

 precipitation_mm
Mean: 60.13260869565213
Mode: [26.1, 47.5, 41.1]
Median: 53.1
Variance: 771.0053957231916
Standard deviation: 27.76698391477172
IQR: 40.75

City: Philadelphia
Frequency: 552

 temperature_celsius
Mean: 8.371431159420295
Mode: [17.56]
Median: 8.655000000000001
Variance: 117.00384059779324
Standard deviation: 10.816831356630889
IQR: 21.005

 humidity_percent
Mean: 48.29601449275356
Mode: [51.0]
Median: 48.2
Variance: 88.1399840869038
Standard deviation: 9.38828973172983
IQR: 13.699999999999996

 precipitation_mm
Mean: 59.509420289855065
Mode: [40.4, 46.4]
Median: 51.9
Variance: 767.2237586469919
Standard deviation: 27.69880428189982
IQR: 39.25

City: Phoenix
Frequency: 552

 temperature_celsius
Mean: 13.50211956521739
Mode: [27.46, 12.09, 26.74]
Median: 12.945
Variance: 117.37266755010651
Standard deviation: 10.833866694311247
IQR: 20.365
```

```
 humidity_percent
Mean: 49.64891304347832
Mode: [41.0, 47.8, 47.4]
Median: 49.150000000000006
Variance: 95.21745798153577
Standard deviation: 9.757943327440254
IQR: 13.349999999999994

 precipitation_mm
Mean: 61.00072463768115
Mode: [63.5, 49.1]
Median: 55.849999999999994
Variance: 763.7270049185926
Standard deviation: 27.63561117324154
IQR: 39.05

City: San Antonio
Frequency: 552

 temperature_celsius
Mean: 16.81012681159421
Mode: [2.18]
Median: 16.805
Variance: 115.4190742125647
Standard deviation: 10.743326962005982
IQR: 20.735

 humidity_percent
Mean: 61.81268115942027
Mode: [59.2]
Median: 62.0
Variance: 88.56459388726691
Standard deviation: 9.410876361278312
IQR: 13.5

 precipitation_mm
Mean: 80.39257246376823
Mode: [82.0, 83.9, 65.4, 30.5]
Median: 73.85
Variance: 1307.7043911925603
Standard deviation: 36.16219560801805
IQR: 55.35000000000001

City: San Diego
Frequency: 552

 temperature_celsius
Mean: 14.1866847826087
Mode: [29.52, -0.87, -0.8, 14.7]
Median: 14.219999999999999
Variance: 120.18730314546671
Standard deviation: 10.962996996509062
IQR: 20.82

 humidity_percent
```

```
Mean: 40.57228260869566
Mode: [20.0]
Median: 40.8
Variance: 85.53580022488751
Standard deviation: 9.248556656305215
IQR: 12.700000000000003

 precipitation_mm
Mean: 59.71467391304344
Mode: [44.5]
Median: 54.150000000000006
Variance: 743.883250710171
Standard deviation: 27.274223191690922
IQR: 40.64999999999999

City: San Jose
Frequency: 552

 temperature_celsius
Mean: 10.5841304347826
Mode: [-2.77, -1.59]
Median: 10.895
Variance: 117.7295506036455
Standard deviation: 10.850324907745643
IQR: 20.575

 humidity_percent
Mean: 35.9494565217391
Mode: [20.0]
Median: 35.849999999999994
Variance: 90.20653327941292
Standard deviation: 9.497712002341034
IQR: 13.199999999999996

 precipitation_mm
Mean: 60.277173913043455
Mode: [30.6]
Median: 53.849999999999994
Variance: 776.3841967174301
Standard deviation: 27.86367163023262
IQR: 41.45

City: São Paulo
Frequency: 552

 temperature_celsius
Mean: 21.60260869565218
Mode: [21.17, 33.98, 34.7, 35.62]
Median: 21.79
Variance: 116.34985452536903
Standard deviation: 10.786558975195428
IQR: 20.425

 humidity_percent
Mean: 38.638586956521756
Mode: [20.0]
```

```
Median: 38.95
Variance: 74.95852649333233
Standard deviation: 8.657859232704833
IQR: 12.049999999999997

 precipitation_mm
Mean: 78.86322463768109
Mode: [58.7]
Median: 69.1
Variance: 1415.7654146282118
Standard deviation: 37.62665829738554
IQR: 53.85000000000001

City: Tokyo
Frequency: 552

 temperature_celsius
Mean: 12.789076086956522
Mode: [22.69, 1.19, 6.36, 0.23]
Median: 12.815
Variance: 116.16382001597887
Standard deviation: 10.777932084401852
IQR: 19.955000000000002

 humidity_percent
Mean: 42.47989130434786
Mode: [47.3, 43.7, 47.8, 45.0]
Median: 42.8
Variance: 93.15126459796411
Standard deviation: 9.651490278602788
IQR: 13.299999999999997

 precipitation_mm
Mean: 59.095289855072494
Mode: [43.7, 44.7, 23.1]
Median: 53.55
Variance: 767.9268017307148
Standard deviation: 27.711492232117614
IQR: 40.55
```

Inference: Grouping by city allows us to compare the central tendency and spread of climate measurements for each city.

7. Simple conclusion

In this experiment, I calculated frequency, mean, median, mode, variance, standard deviation and interquartile range using the urban climate dataset. I also performed manual statistical calculations using loops and analysed the data city-wise. This experiment helped me understand measures of central tendency and variability in data.

Note on calculations

- Variance and standard deviation in the Python programs use the **sample formula** with `n - 1`.
- Quartiles are calculated by dividing the sorted values into lower and upper halves and finding their medians.
- Small differences may occur if another software uses a different quartile calculation method.